Research paper status (ranking evidence and next XAI work)

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This note records what can be claimed from the current tree, after the paired-OOF late-fusion protocol and its post-hoc calibration stage completed. It is an internal planning document, not a manuscript.

Canonical ranking sources:

- Protocol: PairedOOFLateFusionProtocol.md (`phase8-p0-paired-oof-late-fusion-v2`)
- Aggregate metrics: HybridModelResultReview.md
- Gate report: `reports/phase8_p0_paired_oof_late_fusion_gate_score.json`
- Calibration report: `reports/phase8_p0_paired_oof_calibration.json` (`phase8-p0-paired-oof-calibration-v1`, OAR `22662`)
- XAI design (unimplemented explainers): XAIInterpretabilityResearch.md
- Public status: README.md, ARCHITECTURE.md

Observed prescriptions remain historical labels, not optimal treatment. Ranking promotion is an offline research decision. It does not authorize clinical use.

Locked ranking result

Protocol `phase8-p0-paired-oof-late-fusion-v2` finished protected OOF, selection, refit, and the one-shot MIMIC-test gate `mimiciv-test-paired-oof-v1`. Selected pair: GNN variant `full`, $\alpha = 0.25$. Gate decision: `qualified=true`, `final_scoring_authorized=true`.

Setting	Reference	Δ NDCG@10	Δ MRR@10	Δ Hit@10	Role
Asymmetric hybrid (validation)	Frozen Transformer	+0.00484	+0.0039	+0.0048	Historical negative control; failed +0.005
Paired OOF (train selection)	Best single branch (Transformer)	+0.0092	+0.0080	+0.0072	Leakage-safe selection only
Paired gate (MIMIC test)	Frozen Transformer	+0.0131	+0.0122	+0.0116	Passed; hybrid promoted for ranking

Selected blend on test: Transformer NDCG@10 0.4119 → hybrid 0.4250. GNN alone does not beat the Transformer on train OOF (best GNN 0.4068 vs Transformer 0.4137). The `rank_only` partner selected $\alpha = 0.00$; `full` selected $\alpha = 0.25$.

Ranking promotion is not probability calibration. The ranking-gate Brier (hybrid 0.262 vs Transformer 0.210) and 10-bin ECE (0.420 vs 0.337) are **uncalibrated** score disclosures on the same test rows: they describe raw rank scores treated as probabilities, not the frozen mapping. Post-hoc calibration (`phase8-p0-paired-oof-calibration-v1`) completed on OAR `22662`. Both branches froze affine sigmoid on paired train OOF. Cross-fitted Brier ≈ 0.069 and 10-bin ECE ≈ 0.001 (hybrid 0.0688 / 0.0014; Transformer 0.0687 / 0.0013). Ranking `rank_score` stays frozen. Applying that mapping to the consumed MIMIC-test gate would be retrospective only.

Disclosure: the frozen Transformer had already been scored on MIMIC test in the broader program. The paired-fusion score is new. A new patient or temporal holdout would isolate the gate more strongly.

What is actually in the repository

Layer	Status	Paper use
MIMIC-IV / eICU data foundation, 24h feature cutoff, (24h, 48h] labels, patient splits	Implemented	Case-study protocol
Popularity, linear, XGBoost baselines	Implemented; Stage 1 winner is `xgboost_rank_ndcg_oof_late_fusion` (val NDCG@10 ≈ 0.395)	Required controls
Transformer patient/context ranker	Implemented and trained	Sequential view
Patient-subgraph R-GCN, 6 pre-registered ablations	Implemented; standalone GNN beat graph-only XGBoost (0.4015 vs 0.2985 NDCG@10)	Relational view
Late fusion `(1-`α`) z(T) + `α` z(G)`	Implemented	Fusion operator
Asymmetric fusion (GNN OOF + full-train Transformer)	Completed; +0.00484 NDCG@10, failed the +0.005 gate	Negative control
Paired-OOF fusion protocol	Completed; `full`, $\alpha = 0.25$; test gate passed (+0.0131 NDCG@10)	Measured ranking identification result
Paired-OOF probability calibration	Completed (OAR `22662`); both branches affine sigmoid; train-OOF Brier ≈ 0.069, ECE ≈ 0.001	Separate from ranking; test Brier/ECE remain uncalibrated disclosures
Evidence ledger (`pipeline.explainability`)	Typed schema + conservation checks + tests	Next implementation; explainers not built

Layer	Status	Paper use
Attribution, GNN paths, rules, LLM verbalization	Not implemented	Cannot claim
Conversational understanding	Not implemented	Poster, not a result
eICU ranking performance	Coverage-only	No external-validity claim

The Transformer already includes Stage-1 graph **tabular** summaries. Fusion tests whether **message passing** adds ranking signal on top of those summaries.

Professor feedback and paper object

The clinical product framing in ResearchDetail.md ("explainable conversational medication recommender") is still an engineering system description. A research paper needs a domain-independent question. Medicine is the experimental domain.

Two defensible objects now sit in the repo:

1. **Ranking identification (measured).** Hybrid sequential–relational lift is not identified unless graph construction, preprocessing, and the fusion weight are estimated out of fold. The asymmetric protocol failed the pre-registered bar; the paired protocol passed it on MIMIC test.
2. **Faithful explanation of that fused ranker (design only).** The intended methodological novelty for the next paper is a constraint-aware rank-evidence ledger over the promoted hybrid. The contract exists; IG, subgraph interventions, rule certificates, and language checks do not.

Use the locked hybrid as the **model under explanation**, not as a claim that the conversational CDS system exists.

Why "medical recommender" is engineering here

Medication combination on MIMIC is a crowded line: GAMENet (AAAI 2019), SafeDrug (IJCAI 2021), MICRON, COGNet, MoleRec, 4SDrug, MGRN (2024), EDRMM (2025), DrugDoctor, GraphCare (ICLR 2024), KEHGCN (AAAI 2026), ExpDrug (2025), GraphDiffMed (2026), IngMamba (2026), SafeRx-Agent (2026). Almost all of them predict a **visit-level drug set** and report Jaccard / F1 / PRAUC / DDI rate. They do not solve the temporally cut candidate-ranking task used here, but they occupy the "we recommend medications with graphs and sequences" niche.

Sequence+graph hybrids are also known on the RecSys side: MCLSR, MRGSRec, GSAU, LOOM, and RG-Hybrid (2026), which trains a language encoder and a graph encoder independently and averages logits. Graph leakage during message passing is a known RecSys issue (STAR-GCN, EFLEC). OOF stacking to stop label leakage in graph models is also known (bagged stacking on graphs, 2022).

A new hybrid **architecture** is not a contribution. Two things can be:

- the **identification procedure** (now measured on MIMIC, still needs a second domain and statistical tests);
- the **fused-rank explanation contract** (specified, not evaluated).

The GNN is useful against graph-only XGBoost and is not stronger than the Transformer alone. The event that matters for identification is that **asymmetric stacking almost invented a hybrid, then the promotion rule rejected it**, and **paired OOF selected a heavier GNN**

weight ($\alpha = 0.25$ vs ≈ 0.1) and passed the test gate.

Generic problems sitting inside the current code

These are domain-independent. Medicine is where the project has scale, weak labels, and a typed graph.

1. **Complementary-view ranking under shared entities.** One view is a sequence/context encoder. The other is a heterogeneous graph over the same entities. The question is when fusion is complementarity rather than double-counting.

2. **Transductive graph leakage in ranking.** If coprescription, support counts, vocabularies, and edge weights are fit on all training patients, a held-out patient's neighborhood still contains training labels of related entities. Fold-excluded graph caches are the operational answer.

3. **Asymmetric stacking leakage.** Fitting α on GNN OOF plus a full-train Transformer covariate is not hybrid OOF. Paired OOF is the correction, and the two protocols disagreed on promotion.

4. **Weak observational ranking.** Observed actions are positives; unobserved candidates are not true negatives; popularity and future events leak. The 24h/48h contract is a careful instance of ranking-under-logging.

5. **Faithful explanation of a fused ranker with hard constraints.** RankingSHAP / RankSHAP / ShaRP explain listwise tabular rankers. GNNExplainer-style methods explain graphs. Almost nobody explains the **final fused pairwise margin**, with hard exclusions kept out of the additive decomposition. That design is in XAIInterpretabilityResearch.md. The ledger schema exists. The explainers do not.

Engineering vs scientific contribution

Engineering (appendix / reproducibility): DuckDB pipeline, harmonization, Calculco jobs, candidate catalogs, temperature calibration, checkpoint hashing, fail-closed CLI gates, the clinical UI story.

Scientific (ranking, now partly measured): the estimand for hybrid lift under entity-excluded graphs and paired OOF; the finding that a 0.005-scale fusion decision flipped with that protocol; the six GNN relation ablations; the ranking-with-temporal-cutoff task as a case study.

Scientific (XAI, not yet measured): a conserved pairwise evidence ledger over the promoted hybrid, with clinically valid interventions and hard/soft safety separation.

Not a contribution unless implemented and evaluated: LIME+KG+rules+LLM assembly, conversational CRS, clinician trust. The XAI note already treated that assembly as a straightforward combination.

Research questions and hypotheses

RQ1 (ranking identification; MIMIC evidence exists). Does paired out-of-fold fusion of a sequential ranker and an entity-excluded relational ranker identify a hybrid ranking gain that asymmetric stacking misestimates?

- **H1a (supported on MIMIC, pending stats and a second domain).** Under paired OOF, fused NDCG@10 exceeded the better single view by +0.0092 on train OOF and +0.0131 on the MIMIC-test gate, with MRR@10 and Hit@10 also improved.

- **H1b (supported as a protocol contrast, not a causal proof).** Asymmetric stacking selected $\alpha \approx 0.1$ and failed validation promotion (+0.00484). Paired OOF selected $\alpha = 0.25$ and passed the test gate. The Transformer was not wholly unseen on that test split.

RQ2 (complementarity; supported as an ablation, not as architecture novelty). Is the graph view complementary, or a second copy of catalog rank / coprescription frequency?

- **H2 (directionally supported).** `rank_only` selected $\alpha = 0.00$ (Transformer only). `full` was selected at $\alpha = 0.25$ and beat `no_message_passing` as a fusion partner on train OOF (0.4229 vs 0.4206 NDCG@10). GNN alone still trails the Transformer.

RQ3 (still required for a domain-independent ranking paper). Does the same protocol transfer to a non-clinical sequential+graph ranking dataset?

- **H3.** On that dataset, paired OOF fusion again beats the best single view, and transductive / in-sample stacking overstates or destabilizes α. **Not tested.**

RQ4 (intended XAI contribution; not implemented). Can a conservation-constrained pairwise evidence ledger reconstruct the fused margin more faithfully than independent pointwise attributions?

- **H4.** At matched sparsity, ledger explanations have lower pairwise-margin reconstruction error and higher rank preservation/deletion fidelity than RankingSHAP/RankSHAP on the sequential branch plus GNNExplainer/PGExplainer on the graph, concatenated after the fact.

RQ5 (later than RQ4). Does clause-level checking of generated text against the ledger cut unsupported claims relative to unconstrained LLM rationales? Do not put this in a ranking-only paper, and do not claim it before RQ4 exists.

Mathematical formulation (ranking identification)

Instances. Ranking group g (stay + index condition, or user + query in a public dataset). Candidate slate C_g. Weak labels $y_{g,m} \in \{0, 1\}$.

Sequential view. Events E_g before cutoff t_{pred}, static context x_g:

$$\mathbf{h}_g = f_\theta(x_g, E_g), \qquad t_{g,m} = s_\theta(\mathbf{h}_g, c_m).$$

Loss: multi-positive listwise softmax plus auxiliary BCE (`pipeline.neural_training.losses`).

Relational view. Heterogeneous graph $\mathcal{G}^{(-k)}$ fit **without** fold-k entities. Patient (or user) subgraph $G_g \subset \mathcal{G}^{(-k)}$. R-GCN-style update (implemented grouped operator):

$$\mathbf{h}_v^{(l+1)} = \mathrm{LN}\Big(\mathbf{h}_v^{(l)} + \sigma\Big(\sum_{r \in \mathcal{R}} \gamma_r W_r \sum_{u \in \mathcal{N}_r(v)} w_{uv}\, \mathbf{h}_u^{(l)}\Big)\Big),$$

with $\gamma_r = \sigma(\beta_r)$. Candidate score $g_{g,m} = q_\phi(\mathbf{h}_{\text{query}}, \mathbf{h}_m, \mathbf{h}_{\text{ctx}})$.

Fusion. Within-group z-scores over finite, unmasked candidates:

$$z(s)_{g,m} = \frac{s_{g,m} - \mu_g(s)}{\sigma_g(s) + \varepsilon}, \qquad u_{g,m} = (1 - \alpha)\, z(t)_{g,m} + \alpha\, z(g)_{g,m}.$$

Locked selection: $\alpha = 0.25$, variant `full`.

Estimand. Let $\mathcal{M}^* = \arg\max\{\text{NDCG@10}(t), \text{NDCG@10}(g)\}$. Hybrid contribution is

$$\Delta = \text{NDCG@10}(u_{\hat{\alpha}}) - \text{NDCG@10}(\mathcal{M}^*),$$

where $(\text{variant}, \hat{\alpha})$ is chosen on **paired train OOF only**, then evaluated once on a frozen gate.

Identification conditions: patient/user-level splits; temporal cutoff; train-only candidate catalog; fold-excluded graph statistics; no test edges in message passing; α not tuned on the gate split.

If hard safety rules are added, keep them **outside** u as a certificate $h_m \in \{0, 1, \text{unknown}\}$. That belongs with RQ4.

Experiments

Case study A (done for ranking metrics, not for stats). MIMIC-IV ICU stay, condition-scoped candidate ranking, NDCG@10 primary, MRR@10 and Hit@10 guards. Labels = observed starts in (24h, 48h]. Imitation of historical prescribing, not optimal therapy.

Case study B (still missing for a generic ranking paper). One public sequential+graph ranking dataset (for example Amazon-Books or LastFM-17K with item–item / KG edges). Same protocol: sequence encoder, relational encoder, paired OOF fusion, transductive vs entity-excluded graphs.

Baselines

- Sequential only (Transformer; SASRec/BERT4Rec on dataset B)
- Graph only (GNN; LightGCN or R-GCN on dataset B)
- XGBoost / LambdaMART on tabular features
- Equal-weight late fusion (RG-Hybrid style, $\alpha = 0.5$)
- In-sample stacking (full-train sequential + GNN OOF) — the failed asymmetric protocol, already run on MIMIC validation
- Transductive graph vs fold-excluded
- GAMENet/SafeDrug only if their visit-level set task is reimplemented. Do not compare Jaccard on their task to NDCG on this one.

Ablations (already scored on train OOF): `full`, `rank_only`, `no_message_passing`, `no_condition_medication`, `no_dense_lab_vital`, `no_lab_vital_intervention`. Late vs residual fusion was compared on the older asymmetric run (late won).

Metrics: NDCG@10, MRR@10, Hit@10; train-OOF calibrated Brier/ECE as a separate probability axis (ranking-gate Brier/ECE disclose uncalibrated scores); cold-start slice; α selected vs $\alpha = 0/0.5/1$.

Statistics still missing: patient-clustered bootstrap CIs on Δ; paired test on per-group NDCG. The +0.005 rule is a pre-registered **decision policy**, not a theorem. Report the interval as well. Extra seeds are stability analyses, not fishing.

What not to claim: clinical usefulness, eICU SOTA, conversational quality, explanation faithfulness, or that the ranking-gate Brier/ECE are calibrated probabilities (those figures disclose uncalibrated scores; the train-OOF affine-sigmoid fit is a separate probability layer).

How medicine stays in the paper without becoming the paper

Write the method with users, items, entities, views, graphs, and OOF.

Section 5.1: MIMIC-IV is experimental domain 1, weakly labeled, temporally cut ranking over a heterogeneous clinical graph.

Section 5.2 (if RQ3 is in scope): public RecSys dataset, same method.

If the manuscript is the XAI paper, the hybrid is a frozen system under explanation. MIMIC ranking numbers support that the object being explained is the promoted fused ranker, not a Transformer-only model and not the failed asymmetric blend.

What is still missing

Done (do not list as pending): paired-OOF recovery, freeze of (`full`, 0.25), one-shot MIMIC-test ranking gate, protected paired-OOF calibration fit (OAR `22662`; both branches affine sigmoid).

Still needed if the paper is ranking identification:

1. Patient-clustered bootstrap CIs and a paired test on NDCG@10 deltas.
2. Explicit comparison table that also includes $\alpha = 0.5$ and a transductive-graph control (asymmetric vs paired is already available).
3. One non-clinical sequential+graph dataset (RQ3).
4. Rewrite ResearchDetail.md so the repo question matches the manuscript.

If the paper reports probabilities: use the frozen affine-sigmoid mapping (`probability_observed_prescription`) and keep ranking claims on `rank_score`. Do not cite the ranking-gate Brier/ECE as calibrated. A retrospective test application of that mapping is optional and is not a new confirmatory split.

Should have:

5. Cold-start and low-support slices.
6. eICU performance only if mapping produces in-catalog positives.
7. Runtime/cost of fold-excluded graphs, as a limitation.
8. A stronger unseen gate (new patients or a later time cut) if reviewers reject reuse of MIMIC test for the Transformer.

Needed before an XAI paper (current design/implementation target):

9. Grouped IG/occlusion on the Transformer branch.
10. Contrastive subgraph masks with necessity/sufficiency on the GNN.
11. Exact fusion decomposition on a fixed slate; conservation residual.
12. RankingSHAP/RankSHAP and GNNExplainer/PGExplainer baselines.
13. Optional small expert study if the venue is clinical.

Not needed for either paper yet: LLM conversation, production LIME, clinician deployment, DDI-aware decoding, notes.

Papers to compare against, and venues

Protocol / fusion / leakage: RG-Hybrid (2026); LOOM / MCLSR sequence–graph hybrids; STAR-GCN; EFLEC (SIGIR 2022); bagged graph stacking (2022); LightGCN.

EHR graphs / patient subgraphs: GraphCare (ICLR 2024); R-GCN (Schlichtkrull et al.).

Medication rec, related work with a different estimand: GAMENet, SafeDrug, MICRON, MoleRec, COGNet, MGRN, KEHGCN, DrugDoctor.

XAI: RankingSHAP (SIGIR 2025), RankSHAP, ShaRP, GNNExplainer, PGExplainer, KAPER.

Venues for a ranking-identification paper: ACM RecSys, CIKM, WSDM, The Web Conference (short), ACM TOIS. KDD only if dataset B and the statistical tests are clean.

Venues for the XAI paper: SIGIR, FAccT, TOIS, or JAMIA after explainers and a small expert study exist.

Paper structure (professor outline)

1. **Title / abstract.** Name the scientific object. Ranking paper example: *Leakage-safe fusion of sequential and relational rankers.* State H1, the asymmetric miss (+0.00484), and the paired-OOF test pass (+0.0131). XAI paper example: *A constraint-aware rank-evidence ledger for fused sequential–relational rankers.*
2. **Introduction.** Complementary views; shared-entity graph leakage; stacking leakage; EHR ranking as the stress test. For XAI, add that component-wise attributions do not explain the final fused margin.
3. **Related work.** Sequence–graph hybrids; GNN leakage; stacking; medication rec as a crowded application; ranking XAI and GNN explainers.
4. **Method.** Views, fold-excluded graphs, z-score fusion, selection rule. For XAI, the ledger, intervention operators, and hard/soft certificates.
5. **Experiments.** MIMIC ranking table (locked); optional public dataset; ablations; statistical tests. For XAI, fidelity/deletion/conservation and baselines on the frozen hybrid.
6. **Conclusion.** What was identified; labels are logged behavior; pointer to explanation work or remaining faithfulness limits.
7. **References.**

Put Calculco, DuckDB, and clinician-facing diagrams in a reproducibility appendix or a separate technical report.

Practical sequence

1. Treat paired-OOF ranking promotion as locked. Do not rerun the one-shot gate.
2. Agree with the supervisor whether the next manuscript is ranking identification (needs RQ3 + CIs) or the evidence ledger (needs explainers on the frozen hybrid).
3. If ranking: add dataset B and bootstrap CIs before drafting related work.
4. If XAI: implement Stage 0–2 of XAIInterpretabilityResearch.md against the promoted paired fusion (`full`, $\alpha = 0.25$), not the failed asymmetric blend.
5. Rewrite ResearchDetail.md after that choice so the public research question matches the paper.
6. If the paper reports probabilities, use the frozen train-OOF affine-sigmoid mapping and keep it out of the ranking claim. The ranking-gate Brier/ECE stay uncalibrated disclosures.

Legitimate ranking claim (MIMIC, with the test-split disclosure): hybrid sequential–relational ranking gains on this task were not identified by asymmetric stacking and were identified by paired, entity-excluded OOF fusion (`full`, $\alpha = 0.25$, +0.0131 NDCG@10 on the protocol gate).

Legitimate XAI claim (not yet available): none, beyond a typed evidence contract. The next scientific work is to populate and evaluate that ledger on the promoted hybrid.